

Keeping Net Art Alive

VM Controller & Wayback Cache Proxy

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The Problem

"Choose Your Filter!" — Browser Art since the Beginnings of the WWW — ZKM, Feb–Aug 2025

- **24 browser-based artworks** from 1996–2025
- Most require dead technology: IE6, Flash, Java, Pulse 3D
- Original servers no longer exist
- 6+ months of daily operation, minimal staff

Two recurring failures:

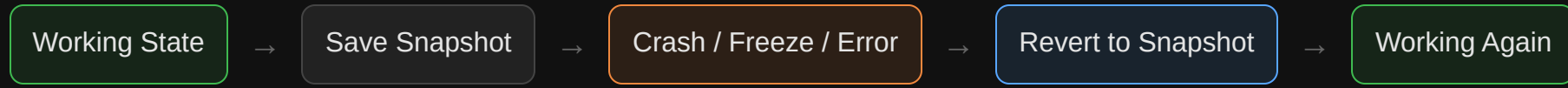
1. **The computer crashes** (Blue Screen of Death, anyone?)
2. **The internet content disappears**



Tool 1: VM Controller

Each artwork runs inside a **virtual machine** — a complete old computer simulated inside a modern one.

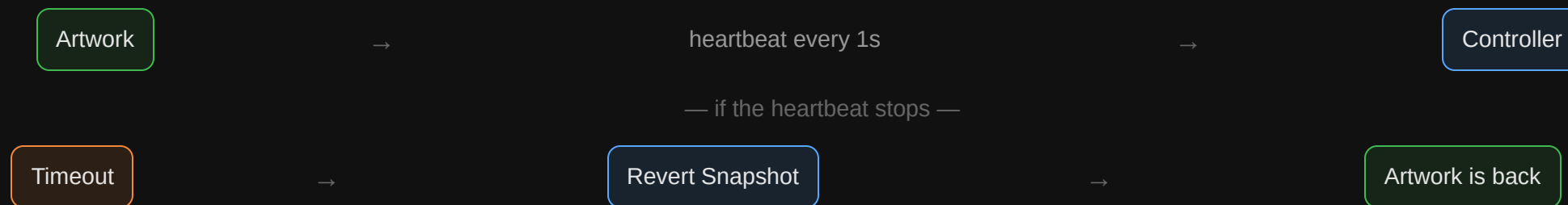
The key idea: take a **snapshot** of a perfectly working state.
When anything goes wrong — *anything* — roll back to that snapshot.



3–5 seconds. No reboot. No repair. Just rewind.

How It Watches

The VM controller monitors each artwork automatically:



- Catches everything: frozen screen, blue screen, crash, power outage
- No human intervention needed
- The artwork doesn't even know it crashed — it just *resumes*

The Three-Layer Model

Layer 3

External Services — live internet, Wayback Machine, replaced APIs

Controlled through the host. Can be eliminated entirely — or managed by the [Wayback Cache Proxy](#).

Layer 2

Host Controller — Python service on the host machine

Manages VM lifecycle, receives guest signals, triggers snapshot reverts, serves replacement services.

Layer 1

VM Guest — the conservation object, never modified after setup

Historical OS + browser + plugins + artwork. Frozen in a snapshot.

Inside the guest, small **Autolt scripts** (compiled to .exe) do the monitoring:

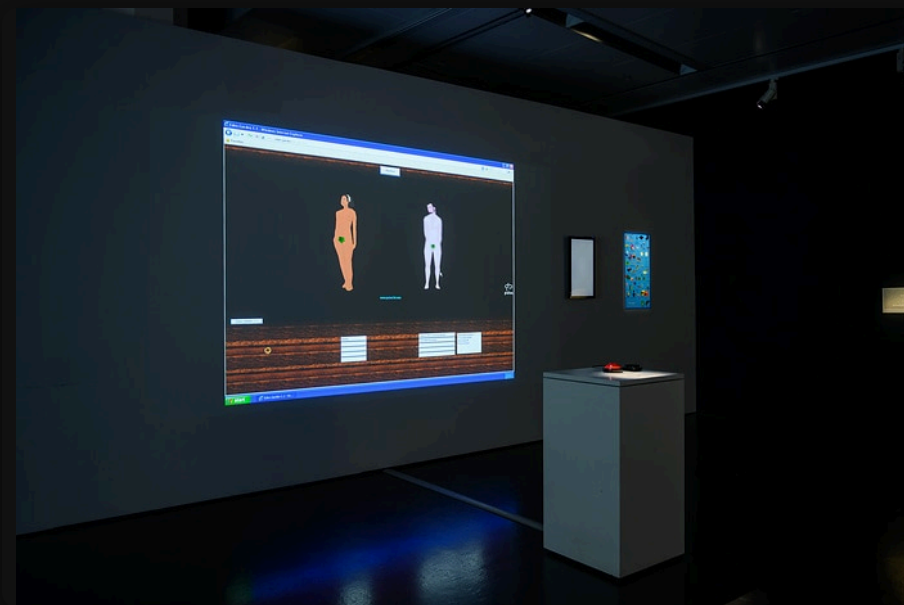
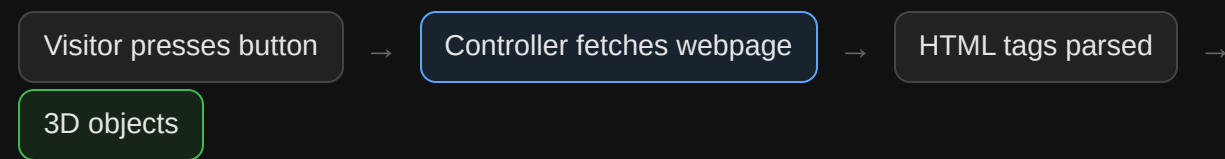
- **Heartbeat** — sends "I'm alive" to the host every second. Stops → revert.
- **Process watchdog** — checks if the artwork's main process is running. Gone → revert.
- **Idle monitor** — no visitor interaction for 12 min → revert to fresh state.
- **Artwork-specific** — e.g. pixel color check to detect a missing UI button.

All communication is simple HTTP between guest and host. **Every failure mode leads back to the same clean snapshot.**

Example: Eden.Garden

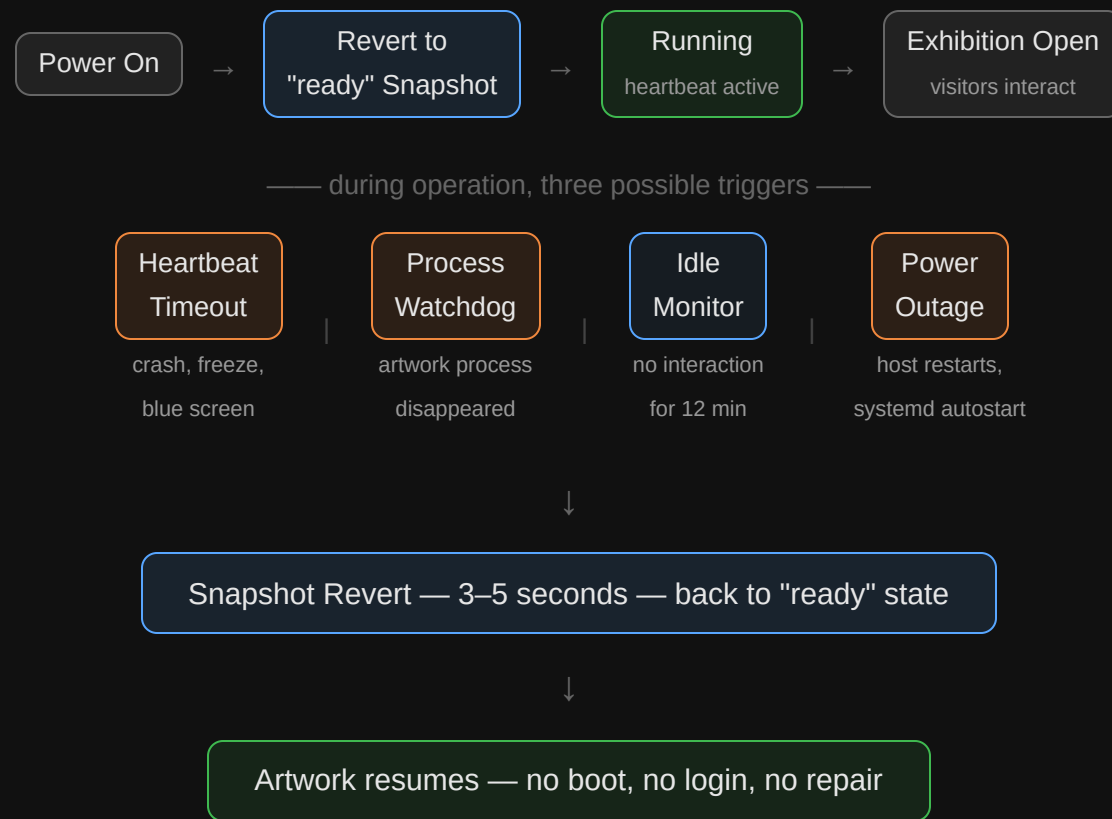
Entropy8Zuper! (Auriea Harvey & Michaël Samyn), 2001

A 3D world built from HTML — required IE + Pulse 3D plugin (company dissolved ~2002).
Not shown for **20+ years**.



VM Lifecycle

The complete state machine for each exhibition VM:



Tool 2: Wayback Cache Proxy

The VM controller handles Layer 1 & 2. But **Layer 3 — external services — can still fail:**

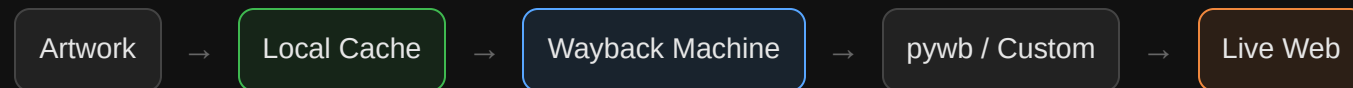
- Many artworks fetch live content from the web
- Original servers are gone (artist pages, APIs, services)
- Even the Internet Archive went down in 2024 (DDoS attack)



The artwork doesn't know the original server is gone. It gets the content it expects, every time.

How the Proxy Works

Configurable backend chain — each artwork can use a different set of sources:



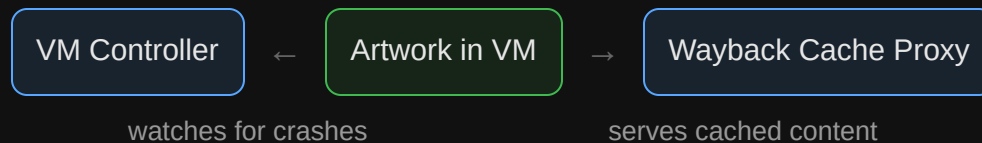
Content sources

- **Local cache** — fastest, works offline
- **Wayback Machine** — Internet Archive CDX API
- **pywb** — replay your own WARC archives
- Custom backends — replaced APIs, search engines, etc.
- **Live web** — last resort fallback

Archival features

- **WARC import** — ingest existing web archives
- **WARC export** — long-term institutional preservation
- Every response auto-cached for next time
- Modem speed simulation for period-accurate experience

Together: Self-Healing Exhibitions



12 VMs

running simultaneously

6+ months

continuous daily operation

0

manual interventions for crashes

2 power outages

all 12 VMs recovered automatically

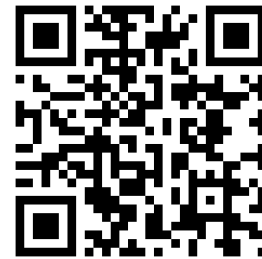
Open Source (MIT)

[exhibition-vm-controller](#)

Snapshot-based self-healing for exhibition VMs

[wayback-cache-proxy](#)

Caching proxy for historical web content



github.com/zkmkarlsruhe

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